SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	3rd Year AI - DS	**Date:**	

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. **Given the following paragraph:**
   "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks:

a) Identify all referring expressions (mentions) and possible antecedents.

S no	Mention	Type	Candidate Antecedents
M1	John	Proper noun (new entity)	—
M2	Mary	Proper noun (new entity)	—
M3	the park	Common noun (new entity)	—
M4	He	Pronoun (masc, sg)	John, Mary
M5	a ball	Common noun (new entity)	—
M6	She	Pronoun (fem, sg)	John, Mary, ball
M7	it	Pronoun (neut, sg)	the park, a ball
M8	The dog	Common noun (new entity)	—
M9	him	Pronoun (masc, sg)	John, Mary, ball, dog
M10	they	Pronoun (pl)	John, Mary, dog, ball
M11	all	Quantifier (reinforces plural)	same set as *they*

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency (prefer recent antecedents)**
- **Semantic compatibility (e.g., animacy and action suitability)**
- **Coherence (maintain consistent entity chains)**

Ans:

1. **Gender/Number agreement** — hard, categorical filter (eliminates candidates of wrong gender/number).
2. **Recency** — among surviving candidates, prefer the most recently mentioned entity.
3. **Semantic compatibility** — animacy and plausibility of the predicate-argument relation (e.g., "play with X", "chase X", "X went home").
4. **Coherence** — the final chain must remain consistent with the discourse read as a whole.

c) Draw a constraint graph or able showing variables, domains, and which constraints eliminate possible links.

Pronoun	Domain (candidates)	Gender/Number filter	Semantic filter	Recency (tie-break)	Resolved to
He (M4)	{John, Mary}	Mary ✗ (fem)	—	—	John
She (M6)	{John, Mary, ball}	John ✗ (masc), ball ✗ (neut)	—	—	Mary
it (M7)	{park, ball}	both neuter, no filter	park ✗ ("play with a park" implausible; ball ✓ playable object)	ball is more recent (Mary "wanted to play")	ball
him (M9)	{John, Mary, ball, dog}	Mary ✗ (fem), ball ✗ (neut)	dog ✗ (same-clause subject cannot corefer with object — *the dog chased him* requires a distinct referent)	John most recent surviving masc. entity	John
they (M10)	{John, Mary, dog, ball}	ball ✗ (inanimate — cannot "go home" as agent)	dog ✓ (animate, accompanies the family)	—	{John, Mary, dog}
all (M11)	reinforces *they*	—	—	—	{John, Mary, dog}

Graph view (nodes = mentions, edges = coreference link, ✗ = eliminated by constraint):

John(M1) —agreement—▶ He(M4) —agreement—▶ him(M9)

Mary(M2) —agreement—▶ She(M6)

ball(M5) —semantic+recency—▶ it(M7)

{John, Mary, dog} —semantic(animacy)—▶ they(M10) —reinforce—▶ all(M11)

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

- **Chain-1 (John):** John → He → him
- **Chain-2 (Mary):** Mary → She
- **Chain-3 (ball):** a ball → it
- **Chain-4 (dog):** the dog
- **Chain-5 (group):** {John, Mary, dog} → they → all

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

Order: Gender/Number agreement → Semantic compatibility → Recency → Coherence.

- *Gender/Number* is applied first because it is a cheap, deterministic, categorical filter that immediately prunes the search space (e.g., removes Mary from "He" candidates).
- *Semantic compatibility* comes next because it uses selectional restrictions (animacy, plausible predicate-argument fit) to remove candidates that survive gender/number but make no sense (e.g., "play with the park").
- *Recency* is used only as a **tie-breaker** among candidates that survive both prior filters, since human/discourse memory favors the most recently activated entity.
- *Coherence* is a final global check — it doesn't eliminate individual candidates but verifies that the whole chain set is internally consistent (e.g., that the same entity isn't split across contradictory chains).

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks:

a) Identify the relevant discourse planning steps needed before generation.

1. **Content selection** — decide to acknowledge the exam and the concentration problem before advising.
2. **Entity/referring** - expression tracking — plan to reuse "exam," "concentrate," and "you" to keep the entity chain coherent.
3. **Discourse relation planning** — choose a Cause-Effect connector or Elaboration.
4. **Speech-act sequencing** — order Advise before Encourage
5. **Lexical/keyword planning** — ensure ≥2 of {focus, break, confident} appear naturally.
6. **Sentence aggregation** — compress content into 2–3 sentences.
7. **Politeness/tone planning** — choose warm, non-judgmental phrasing.
8. **Consistency check** — verify no clause contradicts another.

b) Generate three different possible responses that satisfy all the above constraints.

R1: "I understand exam pressure can make it hard to concentrate, but taking a short break might help you refocus. You're more prepared than you think, so stay confident when you sit down to study."

R2: "It's normal to lose concentration before an important exam, so try taking a short break to clear your mind and regain focus. Trust yourself — you've prepared well, and that should make you feel confident."

R3: "Since you're finding it hard to concentrate for tomorrow's exam, take a five-minute break to reset your mind, and remind yourself that your consistent effort has already built real confidence."

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

Constraint	R1	R2	R3
Entity reuse (exam, concentrate, you)	Partial (exam implied, not named)	Full	Full
Discourse relation (Cause-Effect)	Weak ("but")	Strong ("so")	Strong ("Since...")
≥2 keywords {focus, break, confident}	"break," "confident" (focus only via "refocus")	"break," "focus," "confident" — all 3 explicit	"break," "confidence" (focus missing)
Length (2–3 sentences)	2 ✓	2 ✓	1 long sentence ✗ (borderline)
Logical consistency	✓	✓	✓
Politeness/positive tone	✓	✓	✓

Best:

R2. It explicitly names the exam and reuses "concentrate" and "you," contains all three target keywords with exact matches, uses a clear causal connector ("so") linking the problem to the advice, respects the 2-sentence limit, and closes on an encouraging note — satisfying every constraint with the least ambiguity.

d) Explain how violating any two constraints would affect the quality of the dialog.

- **Violating keyword-inclusion + positive tone:** e.g., "Concentration issues before exams can hurt your score. You should probably study more." — this drops the required vocabulary and turns discouraging, undermining the "Encourage" dialog act and making the response feel judgmental rather than supportive.
- **Violating sentence-length + entity coherence:** a long, rambling response that drifts to unrelated topics (e.g., general life advice not tied back to "exam"/"concentrate") reads as unfocused and impersonal, weakening the discourse coherence the user needs in a moment of stress.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks:

a) Resolve the ambiguous word “bank” and justify using constraints.

Justification via constraints:

- **Selectional restriction:** "flooded" selects for a physical/geographical entity; a financial institution as an organization doesn't literally "flood" (though its building might, that's a strained reading).
- **Collocation cue:** "by the river" is a locative PP that is redundant/uninformative if "bank" already meant a financial institution, but is exactly the disambiguating context for the geographical sense.
- **World knowledge:** "saved by quick action" after a flood evokes flood-control action (sandbagging, reinforcing an embankment), not financial rescue.

b) Write a predicate logic representation of the sentence.

$\exists b\ \exists r\ \exists s\ \exists e1\ \exists e2\ \exists a$

$\text{bank}(b) \wedge \text{river}(r) \wedge \text{located\_by}(b, r) \wedge$

$\text{storm}(s) \wedge$

$\text{flood}(e1) \wedge \text{patient}(e1, b) \wedge \text{cause}(s, e1) \wedge \text{after}(e1, s) \wedge$

$\text{quick\_action}(a) \wedge$

$\text{save}(e2) \wedge \text{patient}(e2, b) \wedge \text{instrument}(e2, a) \wedge$

$\text{Contrast}(e1, e2)$

c) Generate a good target language sentence (in any Indian language or simple English paraphrase) that satisfies all constraints.

Simple English: "The riverbank flooded after the storm, but quick action saved it."

Hindi: "तूफ़ान के बाद नदी का किनारा डूब गया, लेकिन त्वरित कार्रवाई से उसे बचा लिया गया।"

Both preserve all entities (bank/riverbank, river, storm, flood event, save event, instrument) and the Contrast relation, with no information loss and no residual ambiguity.

d) Draw a simple discourse tree (RST-style) showing the rhetorical relation.

[CONTRAST]

(multinuclear)

N

N

"The riverbank flooded

"quick action saved it"

after the storm"

[CAUSE]

(satellite: "after the storm"

modifies nucleus "flooded",

storm = cause of flood)

Relation 1 — Cause: "after the storm" (satellite) → "the bank flooded" (nucleus).

Relation 2 — Contrast (multinuclear): the flooding event vs. the saving event — both are asserted as equally important, contrasted by *but*.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

- **Disambiguation accuracy:** A constraint-based approach uses local syntactic/selectional cues ("by the river," "flooded") to pick the correct sense regardless of corpus frequency; pure statistical MT can default to the more frequent sense (financial "bank") and mistranslate.
- **Relation preservation:** The predicate-logic layer explicitly preserves entity and event relations (who flooded, what caused it, what saved it), whereas phrase-based statistical MT can lose or garble such relations, especially across languages with different syntax.
- **Discourse awareness:** The constraint-based method encodes the Contrast relation explicitly, ensuring the translation keeps the concessive nuance; purely statistical/phrase-level MT often translates clause-by-clause without discourse-level tracking.
- **Interpretability:** Each disambiguation/generation decision can be traced to a specific constraint, making errors easier to diagnose and fix — unlike the opaque probability distributions of pure statistical models.
- **Robustness on rare/ambiguous cases:** Constraint-based reasoning doesn't depend on having seen enough training examples of this exact ambiguous construction, so it generalizes better to low-frequency or novel ambiguous sentences.
- **Trade-off:** constraint-based systems are typically less fluent and harder to scale across a whole language than statistical/neural MT — the advantage above is specific to precision-critical, ambiguity-heavy cases like this one.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________